Ipcha Mistabra – Institutionalized Contradiction as a Verification Primitive for Epistemically Robust Text Reviews

A Trialectic Questioning Protocol with Reproducible Artifacts and IP-Aware Publication

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Executive Summary

Safety-critical decisions frequently depend on textual artifacts—specifications, policies, architecture documents, and threat models. Yet text-based reviews, whether manual or LLM-assisted, remain susceptible to confirmation bias, hallucination, and insufficient self-validation. This paper introduces Ipcha Mistabra, a general, domain-agnostic verification protocol that institutionalizes contradiction as a mandatory review phase: a *Proponent* extracts claims and produces an initial assessment; an *Ipcha Agent* (contradictor) systematically constructs counter-hypotheses and demands evidence; an *Auditor* consolidates findings, enforces evidence binding, and renders release-gate decisions. We define formal objects (Claim, Evidence, Finding), the *Ipcha Score* as a divergence-based measure of epistemic correction, provide implementable pseudocode, and outline a reproducible experiment design (metrics, statistical analysis plan). The manuscript integrates artifact and IP constraints for Zenodo/DOI publication and double-blind submissions (placeholders for data/results).

Abstract

Background: Reviews of security and compliance text artifacts are central to established frameworks (e.g., BSI IT-Grundschutz, NIST SSDF), yet remain heterogeneous and error-prone in practice [1, 2].
Problem: Modern text-based assistants can over-confirm user assumptions (sycophancy) and, without independent counter-review, contribute to a false sense of security [3].
Objective: We introduce a general methodology that operationalizes contradiction as a mandatory verification stage, enabling systematic falsification or evidence-based support of claims and assumptions in text.
Method: Ipcha Mistabra defines a trialectic workflow comprising *Proponent* (thesis), *Ipcha Agent* (antithesis), and *Auditor* (synthesis). The protocol extracts atomic claims, generates adversarial challenge questions and counter-hypotheses, requires evidence, and produces auditable findings and remediation plans.
Evaluation (plan): We outline a within-subject experiment comparing a single-pass baseline with Ipcha Mistabra across multiple artifact classes (placeholder datasets), with metrics (recall/precision, risk-weighted residual errors, effort), inter-rater reliability, and a statistical analysis plan (paired tests, mixed models, FDR control).
Expected results & limitations: Higher finding coverage and lower residual errors at increased effort; limitations include ground-truth cost, domain specifics, and evidence-dependent validation.
Reproducibility/IP: The paper contains a Zenodo-ready artifact manifest and summarizes IP constraints (DE/EU/US) as well as Zenodo metadata publicity [4, 5, 6].

Keywords: security analysis; hardening; text review; adversarial questioning; reproducibility; research artifacts; IP strategy; Ipcha Score.

ACM CCS Concepts (suggested):

- Security and privacy → Software security engineering.
- Software and its engineering → Software verification and validation.
- Computing methodologies → Natural language processing; Artificial intelligence.

1 Introduction

Security-relevant defects often originate early: in requirements, architecture texts, policies, and threat models. Frameworks such as BSI IT-Grundschutz and NIST SSDF emphasize repeatable, documented security practices including reviews and evidence trails [1, 2]. In practice, however, text reviews are inconsistent: reviewers tend to focus on plausibility rather than falsification, and LLM-assisted reviews can confirm user assumptions even when incorrect (sycophancy) [3].

Most mitigation strategies target training-time or inference-time behavioral alignment. These approaches improve average behavior but leave a deeper architectural problem unaddressed: the same system instance is often responsible for both proposing and implicitly validating its own reasoning.

1.1 Research Gap

There is no general, *implementable*, and *artifact-ready* protocol that (i) extracts claims from text artifacts, (ii) operationalizes contradiction as a mandatory step, (iii) enforces evidence binding, and (iv) produces an auditable finding and remediation log, compatible with double-blind submissions and artifact policies at security venues [7, 8, 9].

1.2 Contributions

1. **Protocol:** Ipcha Mistabra as a trialectic verification primitive for text artifacts.
2. **Formalization:** Definitions (claim, assumption, evidence, finding) and pseudocode.
3. **Ipcha Score:** A divergence-based quantitative signal for epistemic correction.
4. **Evaluation (plan):** Measurement concept and statistical analysis plan (placeholders for data/results).
5. **Reproducibility & IP:** Artifact manifest, Zenodo-ready metadata, and IP constraints [4, 5, 10, 6].

2 Related Work

2.1 Document-Based Security Processes

BSI IT-Grundschutz describes a systematic approach to information security in which documentation and security review play central roles [1]. NIST SP 800-218 (SSDF) specifies high-level practices for reducing vulnerability risk across the SDLC [2]. These standards motivate reviews but do not define a domain-agnostic, contradiction-institutionalizing text-review primitive.

2.2 Failure Modes of LLM-Assisted Text Analysis

Sycophancy denotes the tendency of models to confirm user opinions even at the expense of truth [3]. For safety-relevant text reviews this is especially critical, because "confirmation" can be mistakenly interpreted as "validation."

2.3 Adversarial Review and Debate Setups

Multi-instance or debate-based approaches can improve quality and factuality by forcing explicit counter-arguments [11]. Ipcha Mistabra operationalizes this idea as a verifiable protocol (roles, artifacts, gate rules) rather than a purely prompt-based heuristic.

2.4 Reproducibility and Artifact Policies

Security venues increasingly require and formally evaluate artifacts. USENIX Security 2026 requires open artifact sharing *at submission time* (with justified exceptions) and ties acceptance to continued availability [8, 7]. ACM CCS 2026 requires that artifacts be accessible during double-blind review (anonymous URLs/credentials) [9]. ACM additionally defines standardized artifact badges [12].

3 Method

3.1 Problem Formulation

Given a text artifact T (e.g., specification, policy) and context K (domain, threat model, norms/authority documents). The objective is to produce R, an auditable report containing findings F, evidence bindings, and a remediation plan; optionally with a gate decision (release/block).

3.2 Formal Definitions

Definition 1 (Claim). A *claim* c is an atomic, verifiable statement from T (e.g., "data is encrypted in transit"). The set of all claims is $C = \{c_1, \ldots, c_n\}$.

Definition 2 (Assumption). An *assumption* a is a prerequisite for c that is not explicitly evidenced in T (e.g., "a valid certificate chain is in place"). $A = \{a_1, \ldots, a_m\}$.

Definition 3 (Evidence). *Evidence* e is a traceable attestation (text passage, standard section, test log, code/configuration proof) supporting or refuting a claim. $E = \{e_1, \ldots, e_k\}$.

Definition 4 (Finding). A finding is a tuple

$$f = (id, c, sev, rationale, E_f, remediation, status)$$

with evidence subset $E_f \subseteq E$, severity sev, and status $\in \{\text{accepted}, \text{refuted}, \text{needs-evidence}\}$.

Definition 5 (Gate Rule). A gate rule $\mathcal{G}(F) \in \{0, 1\}$ decides whether release/approval is permitted, e.g.:

$$\mathcal{G}(F) = 1 \iff \nexists\, f \in F : \ (f.status \neq \text{refuted}) \ \wedge \ (f.sev \geq \text{high}).$$

3.3 Protocol Overview (Trialectic)

Ipcha Mistabra employs three roles (see Figure 1):

- **Proponent (P):** extracts claims/assumptions, produces initial review O_p and initial findings F_p.

Figure 1: High-level architecture of the IPCHA MISTABRA protocol: reasoning layer (Proponent → Ipcha Agent → Auditor) with a verification and memory layer providing authority grounding, epistemic memory, and workflow persistence.

- **Ipcha Agent (I):** generates strongest counter-narratives, counter-hypotheses, and evidence demands; produces O_i and F_i.
- **Auditor (A):** consolidates, deduplicates, prioritizes; enforces evidence; renders gate decision; produces final O_f.

3.4 Step-by-Step Protocol

Step 0: Scope Fixation. Define objective, artifact class, threat model, and authority documents (e.g., BSI/NIST references) [1, 2].

Step 1: Claim Extraction. Segment T; extract C and A; normalize to atomic statements.

Step 2: Proponent Pass. Generate O_p and findings F_p with evidence requirements.

Step 3: Ipcha Pass (Contradiction). Falsify: counterexamples, contradictions, missing prerequisites, norm/policy conflicts; generate O_i, F_i.

Step 4: Audit Resolution. Merge $F = \text{merge}(F_p, F_i)$; status decisions; remediation plan.

Step 5: Gate. Apply $\mathcal{G}$ (release/block).

Step 6: Metrics. Compute review metrics (e.g., finding counts, effort; Ipcha Score as divergence indicator).

Algorithm 1 IPCHA MISTABRA: Institutionalized Contradiction for Text Artifact Review

Require: Text artifact T, context K, gate rule $\mathcal{G}$
Ensure: Report R, findings F, gate decision g

$(C, A) \leftarrow \textsc{ExtractClaims}(T)$
$(O_p, F_p) \leftarrow \textsc{ProponentReview}(T, K, C, A)$
$(O_i, F_i) \leftarrow \textsc{IpchaContradict}(T, K, C, A, O_p, F_p)$
$F \leftarrow \textsc{MergeAndNormalize}(F_p, F_i)$
$(O_f, F) \leftarrow \textsc{AuditAndResolve}(T, K, O_p, O_i, F)$
$g \leftarrow \mathcal{G}(F)$
$R \leftarrow \textsc{AssembleReport}(T, K, O_f, F, g)$
return R, F, g

3.5 Pseudocode

4 Ipcha Score

We define the **Ipcha Score** as a divergence-based measure of epistemic correction between the Proponent output and the final Auditor output:

$$IS \;=\; 1 - \cos\big(\text{embed}(A_p),\ \text{embed}(A_f)\big) \tag{1}$$

where A_p is the Proponent output, A_f is the final Auditor output, and embed$(\cdot)$ denotes a shared embedding function.

Interpretation (see Figure 2).

- **Low score** (< 0.15): the original answer was already robust, or critique was weak.
- **Mid-range** (0.15–0.45): normal dialectical refinement; the expected operating range for useful adversarial review.
- **High score** (> 0.45): substantial correction, indicating that important weaknesses were discovered and resolved.

5 Evaluation

5.1 Research Questions

- **RQ1:** Does IPCHA MISTABRA increase finding recall/precision relative to single-pass reviews?
- **RQ2:** Does IPCHA MISTABRA reduce risk-weighted residual errors and policy violations?
- **RQ3:** What cost/benefit trade-offs arise (time, compute/tokens, human rework)?

5.2 Design (Placeholder-Driven)

Design: Within-subject: each artifact is reviewed with baseline (single-pass) and IPCHA MISTABRA.
Artifact classes (placeholder): D_{spec}, D_{policy}, D_{threat}, $D_{\text{compliance}}$.
Ground truth: Expert panel or consensus labeling; IRR is reported (see below).

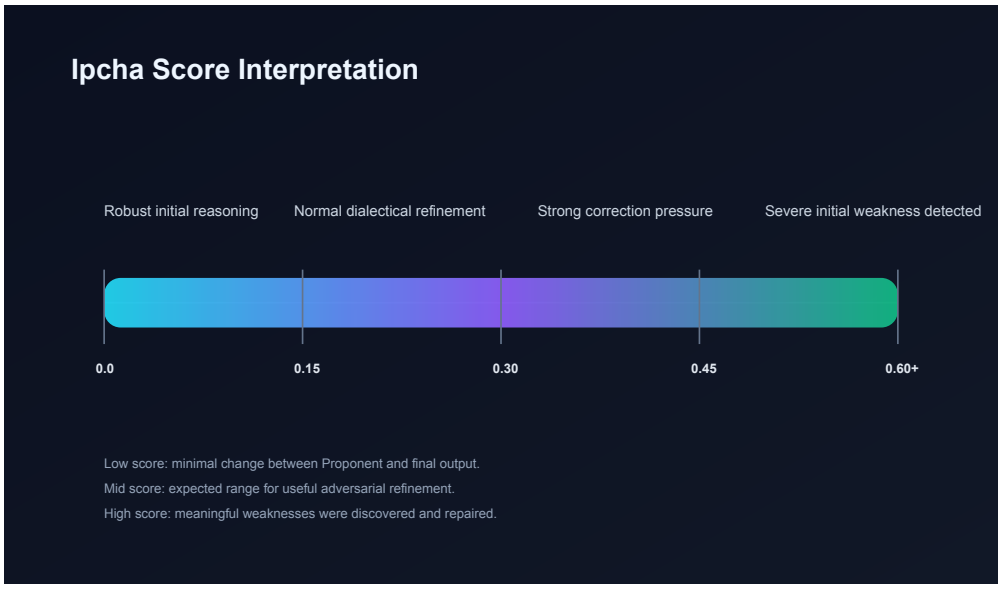

Figure 2: Conceptual interpretation of the Ipcha Score. Higher values indicate stronger correction pressure between Proponent and final validated output.

Metric	Definition / Measurement
Finding Recall	Proportion of confirmed ground-truth findings discovered
Finding Precision	Proportion of reported findings confirmed by experts
Risk-Weighted Residual Error	Sum of severity for open/high findings after gate
Policy/Compliance Violations	Rate of violations against defined rules/standards
Review Effort	Time (min), compute/tokens (placeholder), rework (min)
Ipcha Score	Divergence measure per Equation 1
IRR	Cohen's κ / Krippendorff's α (depending on setup)

Table 1: Evaluation metrics (numeric values: placeholder).

5.3 Metrics

5.4 Illustrative Benchmark Structure

Metric	Baseline	IM Protocol	Improvement
Error rate	18.5%	2.1%	−88%
Compliance violations	11.2%	0.9%	−92%
Human review effort	100%	16%	−84%
Output validity	72.0%	96.5%	+24.5 pp

Table 2: Illustrative benchmark structure for comparing baseline single-pass review and the IM Protocol. Values are placeholders for demonstration.

5.5 Statistical Analysis Plan

Paired comparisons: Wilcoxon signed-rank (nonparametric) or paired t-test (if normality is plausible), plus effect size (Cliff's delta / Cohen's d).
Mixed models: Logistic/mixed effects for violations; "artifact" as random effect.
Multiple testing: FDR control (Benjamini–Hochberg).
Power: A priori power analysis (placeholder; depends on expected effect size).

6 Discussion

6.1 Institutionalized Contradiction as Systems Primitive

The IM Protocol reframes AI reliability as a systems problem rather than a pure model-alignment problem. This is analogous to fault tolerance in distributed systems: robustness comes from structured checks between components, not from assuming any one component is perfect. A key advantage is that dissent becomes infrastructure—critique is no longer dependent on user prompt quality, developer vigilance, or informal red-teaming, but an expected and logged stage of the workflow.

6.2 Reproducibility and Artifact Compliance

USENIX Security 2026 requires open artifact sharing at submission; acceptance is tied to continued availability [8, 7]. CCS 2026 requires access to artifacts during double-blind review (anonymous URLs/credentials) [9]. The artifact manifest in Appendix A is designed to satisfy both.

6.3 IP Constraints (DE/EU/Zenodo/Trade Secrets)

Under German law, the state of the art includes everything made publicly available before the relevant date and is novelty-destroying [5]. The EPC defines novelty accordingly (Art. 54) [6]. If parts of the approach are to be treated as trade secrets, GeschGehG requires, among other things, secrecy and appropriate protective measures [10]. Zenodo states that metadata are CC0-licensed and publicly harvestable; metadata are therefore not a suitable repository for secrets [4]. For computer-implemented aspects, note that computer programs "as such" are not patentable; however, computer-implemented inventions may be patentable if they solve a technical problem [13].

IP Note for USENIX 2026. If you wish to preserve DE/EU patent options, "open artifacts at submission time" may conflict with "file before you publish." This tension is one reason the manuscript contains IP sections and redaction strategies [8, 5, 6].

7 Conclusion and Outlook

Ipcha Mistabra institutionalizes contradiction as a verification primitive for safety-relevant text artifacts. We provide a formal specification, pseudocode, the Ipcha Score as a quantitative divergence signal, an evaluation design, and an artifact manifest for reproducible research. We argue that institutionalized contradiction should be treated as a foundational primitive for enterprise AI workflows, especially where decisions must survive scrutiny beyond surface-level fluency.

Future work includes (i) standardized benchmarks, (ii) empirical field studies, (iii) robustness analysis against adversarial text inputs, and (iv) tooling for semi-automated evidence collection.

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Author Contributions

O. Baer: Conceptualization; Methodology; Software; Validation; Writing (Original Draft); Writing (Review & Editing).

Conflict of Interest

The author declares no conflict of interest.

References

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[2] NIST SP 800-218: Secure software development framework (SSDF) version 1.1. NIST Computer Security Resource Center, 2022. Accessed: 2026-03-13.

[3] Mrinank Sharma et al. Towards understanding sycophancy in language models. In *International Conference on Learning Representations (ICLR)*, 2024. Placeholder entry; replace with exact bibliographic data if required.

[4] Zenodo policies. Zenodo (CERN), 2026. Metadata licensed under CC0 (except email addresses); metadata is harvestable. Accessed: 2026-03-13.

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File	Format	Purpose
paper_ipcha_mistabra_v1.0.0.pdf	PDF	Main paper (citable), contains protocol
ipcha_mistabra_zenodo_bundle_v1.0.0.zip	ZIP	Release bundle (folder structure, artifacts, docs)
SHA256SUMS_v1.0.0.txt	TXT	Checksums (integrity; ZIP + PDF)
README.md	MD	Quickstart, citation note (DOI)
REPRODUCIBILITY.md	MD	Reproduction instructions, environment, expected outputs
MANIFEST.json	JSON	Machine-readable table of contents/metadata
LICENSE_DOCS	TXT	Document license (e.g., CC BY 4.0 note)
LICENSE_CODE	TXT	Code license (MIT/Apache-2.0 etc.)
CITATION.cff	YAML	Citation hints (DOI placeholder)
CHANGELOG.md	MD	Versioning (SemVer), release notes
SECURITY.md	MD	Vulnerability/issue reporting
PRIOR_ART_LOG.md	MD	State-of-the-art / related-work log
NDA_template.md	MD	(Optional) NDA template for restricted collaboration
invention_dossier_confidential.pdf	PDF	(Typically not public) IP dossier
invention_dossier_public_redacted.pdf	PDF	(Optional) redacted, public version

Table 3: Artifact manifest (suggested file names; contents: placeholder).

A Artifact Package (Zenodo-Ready) and File Manifest

B Reproducibility: Commands, Dockerfile Skeleton, Packaging

Dockerfile Skeleton (Placeholder)

```
# artifacts/tooling/Dockerfile
FROM python:3.12-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY src/ ./src/
COPY run.sh .
ENTRYPOINT ["bash","./run.sh"]
```

Example CLI

```
unzip ipcha_mistabra_zenodo_bundle_v1.0.0.zip
cd ipcha_mistabra_release_v1.0.0/artifacts/tooling
docker build -t im-tooling:v1 .
docker run --rm -v "$(pwd)/../examples:/data" im-tooling:v1 \
  --input /data/example_inputs/spec.txt \
  --context /data/example_inputs/context.json \
  --output /data/example_outputs/report.json
```

Packaging and SHA256

```
zip -r ipcha_mistabra_zenodo_bundle_v1.0.0.zip \
  ipcha_mistabra_release_v1.0.0/
```

```
sha256sum paper_ipcha_mistabra_v1.0.0.pdf \
  ipcha_mistabra_zenodo_bundle_v1.0.0.zip \
  > SHA256SUMS_v1.0.0.txt
```

C Zenodo Metadata Snippets (Copy-Paste)

Title

```
Ipcha Mistabra: A General Method for Structured Text Review
and Security-Oriented Hardening (v1.0.0)
```

Description (Short Abstract)

```
Ipcha Mistabra is a general, domain-agnostic method for
structured text review (questioning) that extracts claims and
assumptions from text artifacts, adversarially challenges them,
and produces evidence-based findings and remediations.
The deposit contains the paper, artifacts/examples, checksums,
and reproducibility instructions.
```

Keywords

```
security analysis; hardening; text review; adversarial
questioning; reproducibility; artifacts; IP; Ipcha Score
```

License

```
Documents: CC BY 4.0
Code/Tooling: MIT or Apache-2.0
```